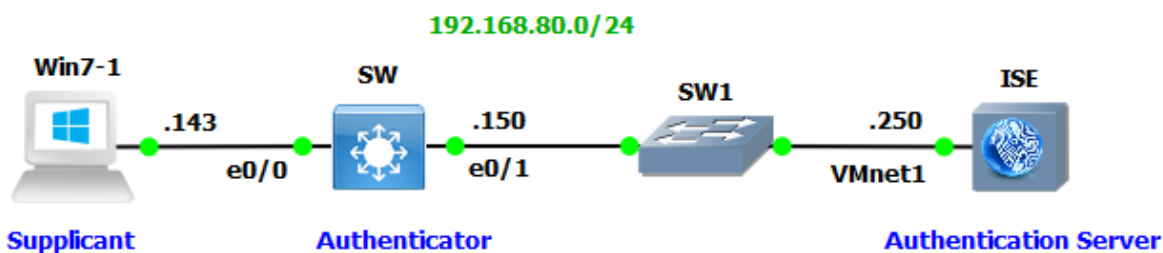


Dot1x LAB:



Older Dot1x Configuration

```
SW(config)#interface vlan 1
SW(config-if)#ip address 192.168.80.150 255.255.255.0
SW(config-if)#no shutdown
SW(config-if)#exit

SW(config)# aaa new-model
SW(config)# username admin password 123
SW(config)# radius-server host 192. 168.80.250 auth-port 1812 acct-port 1813 key test123
SW(config)# aaa authentication dot1x default group radius
SW(config)#aaa authorization network default group radius
SW(config)#aaa accounting dot1x default start-stop group radius
SW(config)#radius-server vsa send
SW(config)#ip device tracking
SW(config)# dot1x system-auth-control

SW(config)# interface e0/0
SW(config-if)# switchport mode access
SW(config-if)# switchport access vlan 1
SW(config-if)# dot1x port-control auto
SW(config-if)# dot1x pae authenticator
SW(config-if)# dot1x host-mode multi-host
SW(config-if)# dot1x reauthentication

SW# show dot1x interface e0/0
SW# show dot1x all
SW# show radius statistics
SW# show dot1x interface e0/0 details
SW# show dot1x interface e0/0 statistics
```

Newer Dot1x Configuration
SW(config)#interface vlan 1
SW(config-if)#ip add 192.168.80.150 255.255.255.0
SW(config-if)#no shutdown
SW(config-if)#exit
SW(config)# aaa new-model
SW(config)# username admin password 123
SW(config)# radius server myserver
SW(config-radius-server)#address ipv4 192.168.80.250
SW(config-radius-server)#key test123
SW(config)# aaa authentication dot1x default group radius
SW(config)#aaa authorization network default group radius
SW(config)#aaa accounting dot1x default start-stop group radius
SW(config)#radius-server vsa send
SW(config)#ip device tracking
SW(config)# dot1x system-auth-control
SW(config)# interface e0/0
SW(config-if)# switchport host
SW(config-if)# switchport access vlan 1
SW(config-if)# authentication port-control auto
SW(config-if)# dot1x pae authenticator
SW(config-if)# authentication host-mode multi-host
SW(config-if)#authentication timer reauthenticate server
SW(config-if)#authentication periodic
SW(config-if)#authentication violation shutdown
SW# show dot1x interface e0/0
SW# show dot1x all

Add Network Device:

Go to **Administration > Network Resources > Network Devices** to add the Device (SW), which will request to authenticate Dot1x Client.



[Network Devices](#)
[Network Device Groups](#)
[Network Device Profiles](#)
[External RADIUS Servers](#)
[RADIUS Server Sequences](#)

[Network devices](#)

Default Device

[Edit](#)
[Add](#)
[Duplicate](#)
[Import](#)
[Export](#)
[Generate PAC](#)
[Delete](#)

Name	IP/Mask	Profile Name	Location
------	---------	--------------	----------

Click on Add button to add Device, Configure **Name** of the device, **IP address** configured.

[Network Devices](#)
[Network Device Groups](#)
[Network Device Profiles](#)
[External RADIUS Servers](#)
[RADIUS Server Sequences](#)

[Network devices](#)

Default Device

[Network Devices List > New Network Device](#)

Network Devices

* Name

Description

* IP Address: /

* Device Profile

Model Name

Software Version

* Network Device Group

Device Type [Set To Default](#)

Location [Set To Default](#)

Scroll down to set Authentication settings. Set Password configured as Server key on Switch device “test123” and save settings.

☒
[RADIUS Authentication Settings](#)

Enable Authentication Settings

Protocol **RADIUS**

* Shared Secret [Show](#)

Enable KeyWrap ☐

* Key Encryption Key [Show](#)

* Message Authenticator Code Key [Show](#)

Key Input Format ☒ ASCII ☐ HEXADECIMAL

CoA Port [Set To Default](#)

[Submit](#)
[Cancel](#)

Add Local Groups & Users:

Now go to **Administrations > Identity Management > Groups > User identity Groups > Add**

The screenshot shows the Cisco Identity Services Engine (ISE) Administration console. The top navigation bar includes 'Home', 'Operations', 'Policy', 'Guest Access', 'Administration', and 'Work Centers'. The 'Administration' menu is expanded, showing 'System', 'Identity Management', 'Network Resources', 'Device Portal Management', 'pxGrid Services', 'Feed Service', and 'Identities'. The 'Identity Management' menu is further expanded, showing 'Groups', 'External Identity Sources', 'Identity Source Sequences', and 'Settings'. The 'Groups' menu is selected, and the 'User Identity Groups' sub-menu is highlighted. The 'Add' button is visible in the top right corner of the 'User Identity Groups' section.

Name	Description
☐ ALL_ACCOUNTS (default)	Default ALL_ACCOUNTS (
☐ Employee	Default Employee User Gr
☐ GROUP_ACCOUNTS (default)	Default GROUP_ACCOUN'
☐ GuestType_Contractor (default)	Identity group mirroring t
☐ GuestType_Daily (default)	Identity group mirroring t
☐ GuestType_Weekly (default)	Identity group mirroring t
☐ OWN_ACCOUNTS (default)	Default OWN_ACCOUNTS

Create New Group For Dot1x Users.

The screenshot shows the 'New User Identity Group' form in the Cisco Identity Services Engine (ISE) Administration console. The form is titled 'User Identity Groups > New User Identity Group'. It has two input fields: 'Name' and 'Description'. The 'Name' field contains 'Dot1x-Group' and the 'Description' field contains 'This is test group for dot1x Users'. There are 'Submit' and 'Cancel' buttons at the bottom of the form.

Now go to **Administrations > Identity Management > Identities > Users** add Set Username and Password for Dot1x Port-based Authentication. Scroll down and Select the user group to **Dot1x-Group** and Save Settings.

The screenshot shows the 'Users' section in the Cisco Identity Services Engine (ISE) Administration console. The top navigation bar includes 'Home', 'Operations', 'Policy', 'Guest Access', 'Administration', and 'Work Centers'. The 'Administration' menu is expanded, showing 'System', 'Identity Management', 'Network Resources', 'Device Portal Management', 'pxGrid Services', 'Feed Service', and 'Identities'. The 'Identities' menu is selected, and the 'Users' sub-menu is highlighted. The 'Add' button is visible in the top right corner of the 'Users' section.

Status	Name	Description	First Name	Last Name
No data				

EndPoints
Users
Latest Manual Network Scan Results

Network Access Users List > **New Network Access User**

Network Access User

* Name **user1**

Status ☒ Enabled

Email

Passwords

Password Re-Enter Password

* Login Password Re-Enter Password

Enable Password

User Groups

Dot1x-Group

Submit Cancel

Allow Protocols:

Now go to **Policy > Policy Elements > Results > Authentication > Allowed Protocols > Default Network Access** and Edit, Check MS-CHAP v2 and Uncheck EAP-TLS from all Locations. Check MS-CHAPv2 and Uncheck EAP-TLS. There will be three Locations of EAP-TLS, which are to be unchecked. Now save the settings.

Cisco Identity Services Engine Home Operations **Policy** Guest Access

Authentication Authorization Profiling Posture Client Provisioning **Policy Elements**

Dictionaries Conditions **Results**

Authentication

Allowed Protocols

Authorization

Profiling

Posture

Client Provisioning

☐ EasyWired

☒ Process Host Lookup

Authentication Protocols

☒ Allow PAP/ASCII

☐ Allow CHAP

☐ Allow MS-CHAPv1

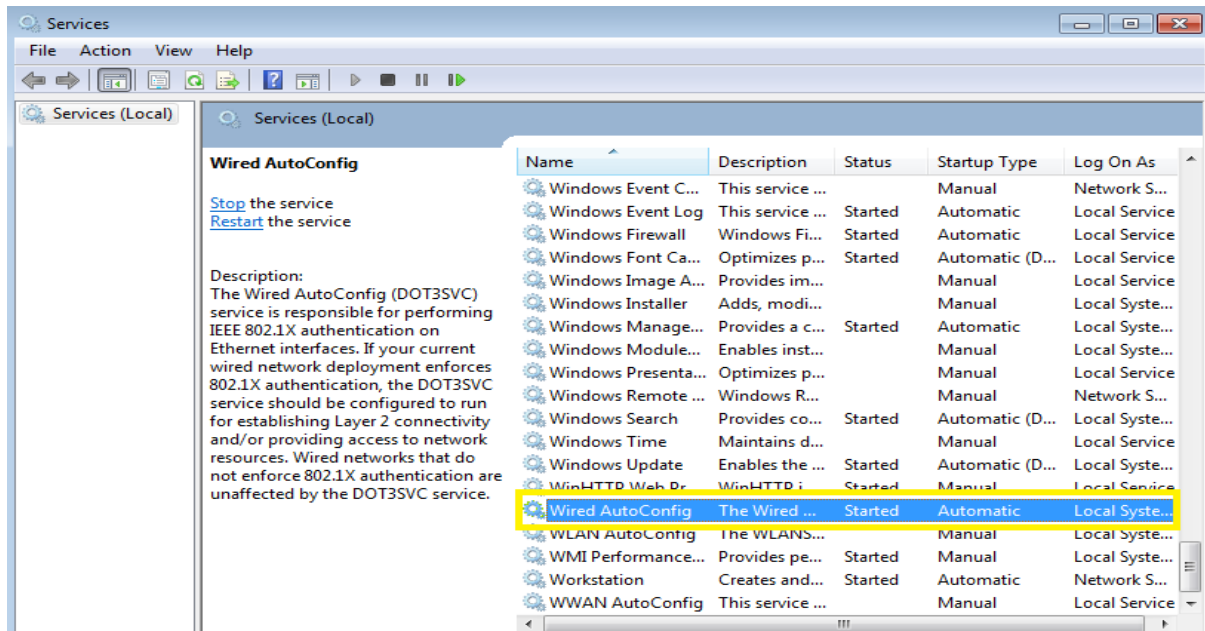
☐ Allow MS-CHAPv2

☒ Allow EAP-MD5

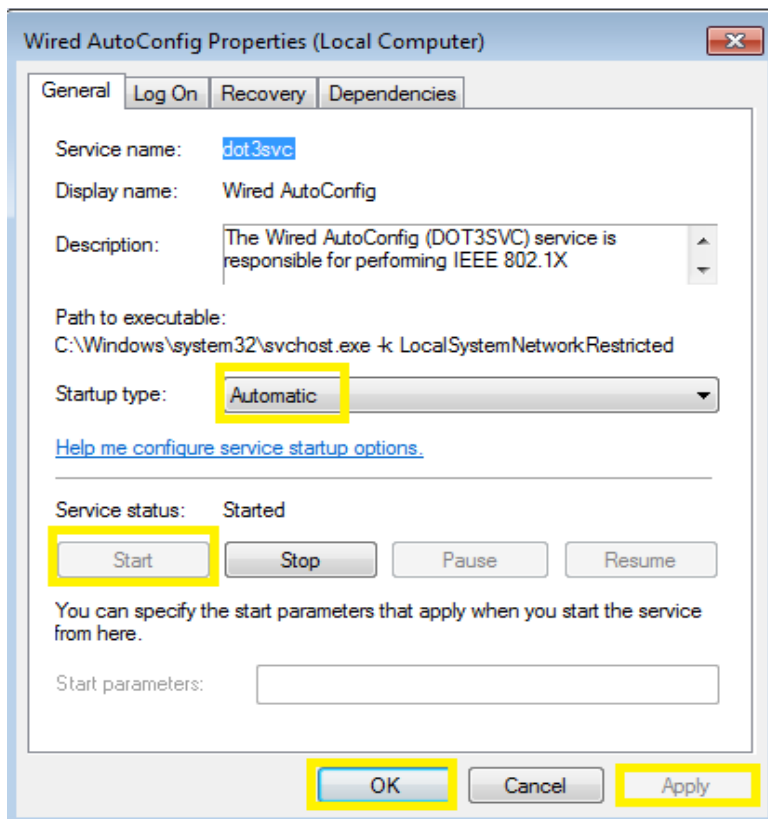
☐ **Allow EAP-TLS**

Dot1x Client Configuration:

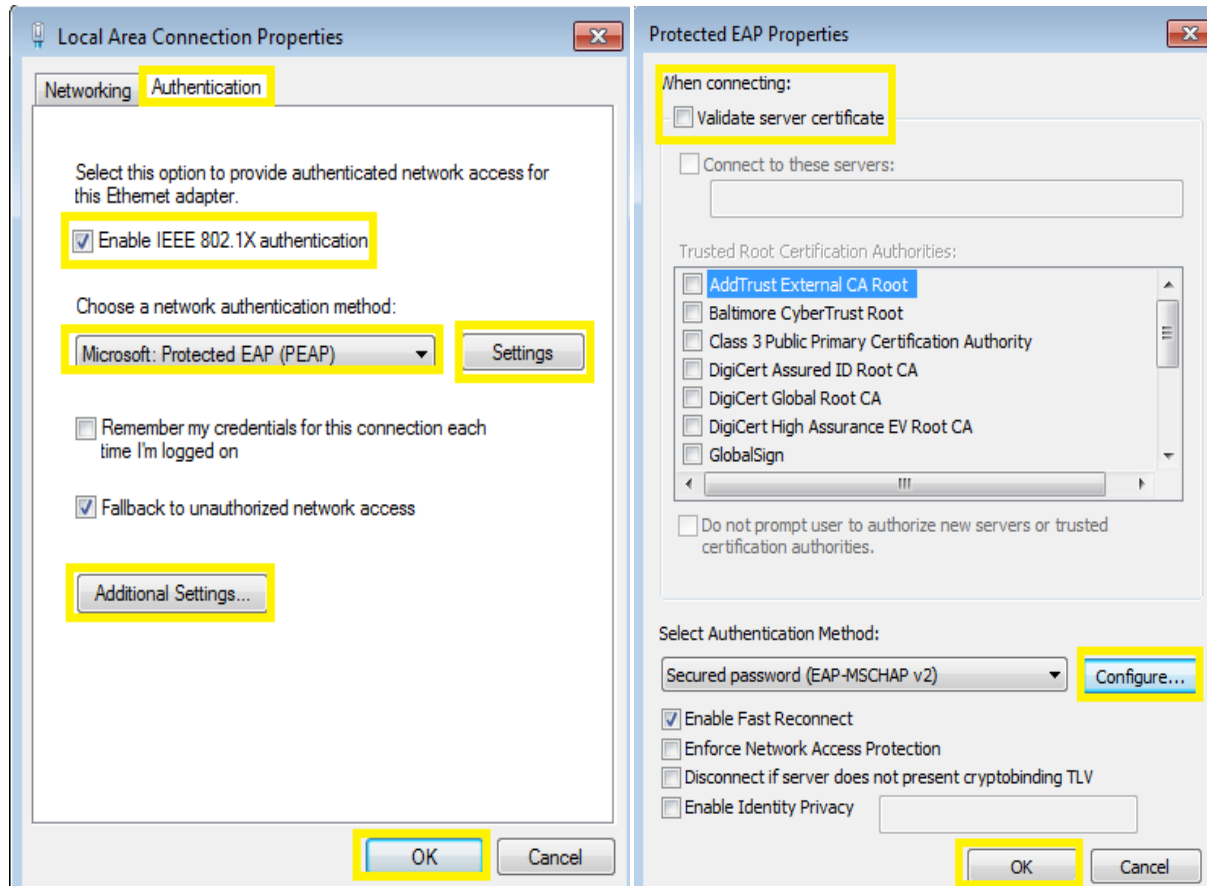
Enable Wired AutoConfig on your Computer. Click the Start button and type **services.msc** into the search box. In the services window locate the service named **Wired AutoConfig**.



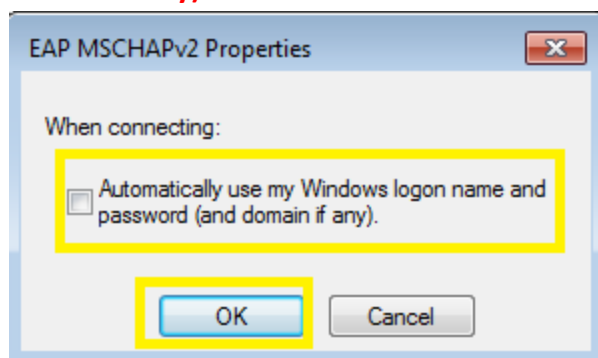
Right click on service & click on **properties**. Select the Startup Type **Automatic** and press **OK**.



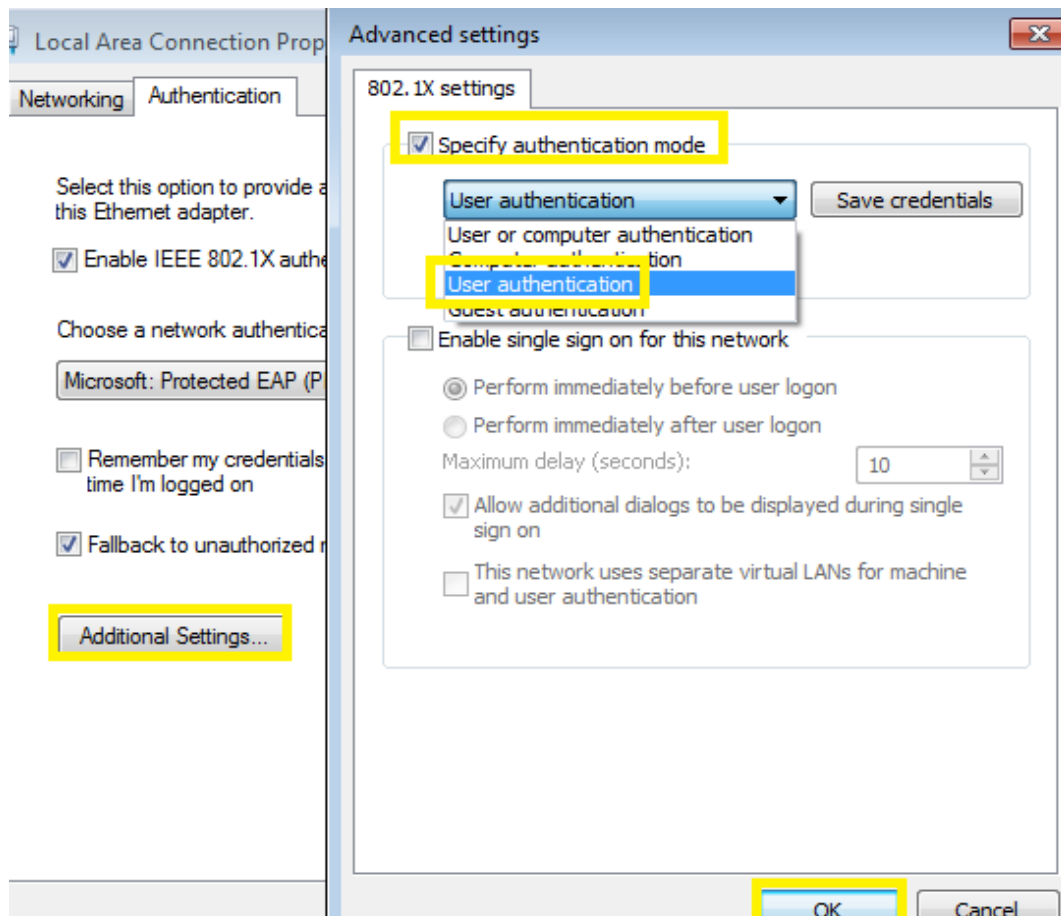
Right click on your network adapter and select **Properties**. Click on the **Authentication** tab and select the option for **IEEE 802.1X authentication**. Choose the network authentication method Microsoft: **Protected EAP (PEAP)** from the drop down: Click **Settings** deselect Validate server Certificate option.



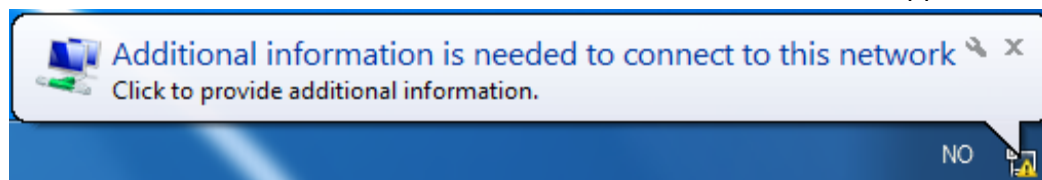
Click **Configure** and deselect **Automatically use my Windows logon name and password (and domain if any)** click "OK" twice to return to the Local Area Connection Properties.



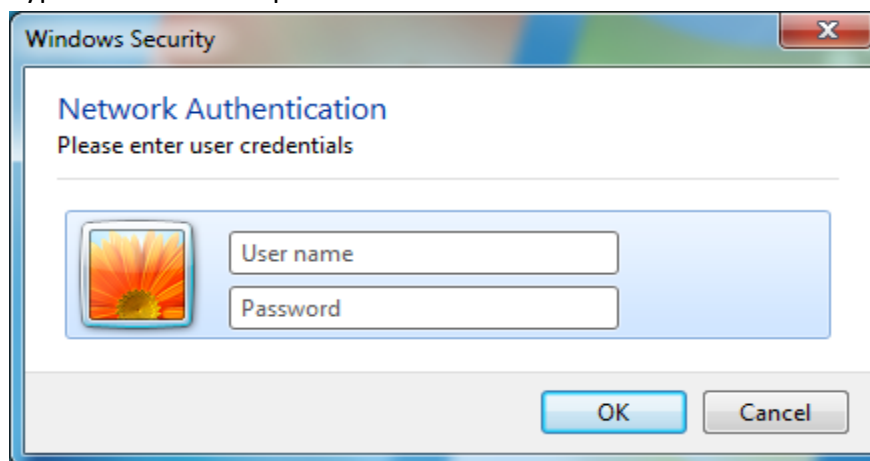
Click **Additional Settings**, select **Specify authentication mode**, and specify **User authentication**. Click "OK" twice.



Local Area Connection bobble. Click it, and Network Authentication appear:



Type username and password to authenticate with Cisco ISE.



Verify:

Navigate to **Operations > RADIUS Livelog**.

Identity Services Engine Home **Operations** Policy Guest Access Administration Work Centers

RADIUS Livelog TACACS Livelog Reports Troubleshoot Adaptive Network Control

Misconfigured Supplicants 0 Misconfigured Network Devices 0 RADIUS Drops 0 Client Stopped Responding 0

Show Live Sessions Add or Remove Columns Refresh Reset Repeat Counts Refresh Never Show Latest 10

Time	Status	Details	Repeat Count	Identity	Endpoint ID	Endpoint Profile	Authentication Policy	Authorization Policy	Authorization I
2019-06-06 19:16:25.901	✗			test	00:0C:29:55:68:E0		Default >> Dot1X >> ...		
2019-06-06 19:15:47.487	✓			user2	00:0C:29:55:68:E0	VMWare-Device	Default >> Dot1X >> ...	Default >> Basic_Aut... PermitAccess	
2019-06-06 19:03:58.512	✓			user1			Default >> Default >>...	Default >> Basic_Aut... PermitAccess	
2019-06-06 19:02:24.953	✓			user1	00:0C:29:55:68:E0	VMWare-Device	Default >> Dot1X >> ...	Default >> Basic_Aut... PermitAccess	

Verification commands on Cisco Switch

SW# show dot1x interface e0/0
SW# show dot1x all
SW# Show authentication sessions
SW# Show authentication sessions interface ethernet 0/0 details